



CHETHAN MYSURU RADHAKRISHNA

M.Sc. in Data and Knowledge Engineering

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portfolio

SKILLS

Python	●	●	●	●	●
R	●	●	●	●	●
SQL	●	●	●	●	●
Postgres	●	●	●	●	●
SPLUNK	●	●	●	●	●
mongoDB	●	●	●	●	●
Node.js	●	●	●	●	●
React.js	●	●	●	●	●
C#	●	●	●	●	●
C++	●	●	●	●	●
Java	●	●	●	●	●

LIBRARIES

PyTorch	Keras	
Tensorflow	scikit-learn	
scikit-image	NLTK	spaCy
gensim	node-red	reSolve

IDES

Visual Studio	PyCharm
Eclipse	Jupyter
Google Colab	

CICD

Bitbucket	Git Lab	TFS
Octopus	Jenkins	Docker

LANGUAGES

English: Professional C1

German: Intermediate B1

ABOUT ME

I am a data science enthusiast with an inclination towards visual exploration of trends in data. I am also a skilled web developer with a collective experience of 4 years. However, I believe in the idea that "What we know is a drop and what we don't know is an ocean".

EXPERIENCE

Big Data Analyst in Project Management Metrics | [Bosch Engineering GmbH](#)

Intern Apr 2021 – Sep 2021 | Work Student Jan 2022 – Dec 2022 Abstatt, Germany

- Worked on the analysis of data originating from software development life cycle using SPLUNK
- Clean-up and modelling of data origination from various sources including sprint planning and requirement management tools like ALM and DOORS
- Developed dashboards for the visualization and dynamic analysis of the full V-Model pertaining to requirement management
- Rolled out the generic real-time dashboards to various projects using CICD tools including bitbucket and Jenkins

Technologies: SPLUNK, Python, SQL

Software Developer | [Metrattec GmbH](#)

Work Student Jun 2019 – Feb 2021 Magdeburg, Germany

- Worked as a full stack web developer and handled API integration with react-redux dashboards
- Worked on real time data using web sockets
- Worked on event sourcing and reSolve(a CQRS framework built using node.js)
- Managed Gitlab pipeline and docker configurations with yaml

Technologies: Reactjs, nodejs, reSolve, web sockets, node-red

Software Engineer | [Happiest Minds Technologies Pvt Ltd](#)

Oct 2016 – Jan 2019 Bengaluru, India

- Contributed as a full stack web developer and worked on ASP.NET MVC applications and Web API services built on .NET framework 4.6.2 and .NET core
- Worked on react.js and redux to implement stateful front-end UI
- Worked exclusively on Health Care domain with the knowledge of online data exchange standards such as HL7
- Worked on AWS EC2, S3 buckets, Simple Mail Service and Lambda(for deploying micro services)

- As a billable resource, interacted with clients to understand requirements and effectively communicated the innovative ideas to solve their problems

Technologies: ASP.NET MVC/WebAPI, C#, Reactjs, nodejs, AWS

EDUCATION

M.Sc. in Data and Knowledge Engineering | [Otto-von-Guericke University](#)

📅 Apr 2019 – March 2024

📍 Magdeburg, Germany

- German GPA: 2.2, CGPA: 8.42

B.E. in Computer Science and Engineering | [Vidyavardhaka College of Engineering](#)

📅 Jun 2012 – Jul 2016

📍 Mysuru, India

- CGPA: 7.68

PROJECTS

Master's Thesis: Exploration of Deep Learning based Vessel Segmentation with Less to No Supervision | [GitHub](#)

📅 July 2023 – March 2024

- The study aims at learning good quality vessel segmentation using 3D patch based unsupervised learning by estimating k-class probability distribution using Attention WNet architecture.
- The research explores the possibility of generating quality vessel segmentation in the absence of manual expert annotations.
- The study also extends the 2D MIP based pipeline to incorporate the weakly annotated MIPs of the 3D volume instead of the full-volume annotations.
- The research also explores the use of weak, noisy labels as supervision using the deformation-aware semi-supervised learning pipeline.
- The study compares the segmentation results of unsupervised, weakly-supervised and semi-supervised approaches and provides a novel performance ranking based on vessel diameters.

Technologies Used: Python, PyTorch, nibabel, scikit-image, cuda, WNet, Deep CNN

Enhancing Vessel Continuity in Deep Learning based Segmentation using Maximum Intensity Projection as Loss | [GitHub](#)

📅 April 2023 – Sep 2024

- Study focuses on challenges pertaining to segmentation of biomedical images by extending "*DS6: Deformation-aware Semi-supervised learning*" published by Dr.Chatterjee et.al., with Maximum Intensity Projection (MIP) comparisons
- Proposed Vessel Segmentation approach was validated using limited annotated samples of high resolution 7Tesla 3D-MRA volumes
- Qualitative analysis of ROI's revealed clear improvement in the continuity of vessel structures perceived by the Deep Learning network after incorporating MIP comparisons as one of the loss terms during training

Technologies Used: Python, PyTorch, nibabel, scikit-image, cuda, docker

Investigating the Challenges of 3D Image Segmentation without Manually Segmented Ground Truth |

 Oct 2022 – Jun 2023

- Research aims at exploring possible benefits of Unsupervised Deep Learning (DL) techniques to reduce the dependency on prior annotations in 3D Image Segmentation
- Research tasks include comparison of quality of segmentations generated using UNet and WNet based architectures with and without limited/weak annotations
- Study extends *Attention WNet* architecture with the similarity and consistency losses proposed in the *Differentiable Feature Clustering (DFC)*
- Approach is validated by performing Pathology Detection(Tumor Segmentation) using 3D MRI volumes obtained using 3Tesla and 7Tesla scanners
- Study compares the resulting segmentations obtained using the proposed approach against non-DL baselines to analyse the efficiency of Deep CNN's in the field of Computer Vision

Technologies Used: Python, PyTorch, nibabel, scikit-image, cuda, WNet, Deep CNN

Interpretable Machine Learning to Understand Participant Evolution in Longitudinal Cohort Study Data |

 Oct 2019 - Mar 2020

- Project aims at identifying evolutionary trends in longitudinal cohort data of a German population in Pomerania
- Study focuses on explainable AI to explore the results of a machine learning model to derive critical factors influencing the diagnosis of Hepatic Steatosis
- Worked on pre-processing, R-API development and visualization of model results using react-redux dashboard

Technologies Used: R, react.js, redux, saga, canvas.js(for interactive graph visualizations)

Information Retrieval System |

 Oct 2019 - Jan 2020

- A console java application which indexes files and directories and allows users to execute boolean queries on them
- Used Apache Lucene to tokenize, index and query text in a directory(nested) or files
- Application aims at achieving disambiguation of search terms and ranking of results based on relevance score

Technologies Used: Apache Lucene, java, maven

Gesture Recognition For Interactive Systems Using Kinect V2 |

 Nov 2015 - Apr 2016

- Developed a WPF application to recognize user gestures using Microsoft Kinect V2 sensor
- Application binds different mouse and keyboard inputs to gestures performed by users and thereby allowing them to interact with the system
- Authentication of users is performed with face recognition algorithm developed using PCA(Principal Component Analysis) to compare grey scale images of faces

Technologies Used: .NET framework 4.5, WPF, Kinect V2 SDK, C#
